🔑 **Answer Key — Teacher Copy · Lesson 7-2**

MSTAR PRACTICE WORKSHEET

Lesson 7-2: Represent Rational Numbers and Their Opposites on the Number Line

Grade 6 Mathematics · Standard 6.NOS.6

These are MSTAR-style questions written for this curriculum to rehearse the format. They are not official Maryland assessment items. MSTAR — Maryland's new state test, first given in spring 2027 — uses the same kinds of questions you see here: selected response, select-all, two-part evidence questions, and written responses.

Part A — Skills Check

One point each.

Question 1 SELECTED RESPONSE — CORRECT: B

Which point is located between -1 and -2 on the number line?

- (A) -2.5
- (B) -1.5
- (C) -0.5
- (D) 1.5

-1.5 is between -1 and -2 because it is 0.5 units to the left of -1 (or 0.5 units to the right of -2).

Question 2 SELECTED RESPONSE — CORRECT: A

Which fraction is equivalent to 0.25?

- (A) 1/4
- (B) 1/2
- (C) 1/3
- (D) 2/5

0.25 means 25 hundredths = 25/100 = 1/4.

Question 3 SELECTED RESPONSE — CORRECT: C

Where is -3/4 located on the number line?

A Between -3 and -4

B Between -1 and -2

C Between 0 and -1

D Between 0 and 1

-3/4 = -0.75, which is between 0 and -1. It is 3/4 of the way from 0 toward -1.

Question 4 SELECTED RESPONSE — CORRECT: B

Which rational number is closest to 0 on the number line?

A -2

B -1/4

C 0.5

D 1.75

|-1/4| = 0.25, |0.5| = 0.5, |-2| = 2, |1.75| = 1.75. Since 0.25 is the smallest absolute value, -1/4 is closest to zero.

Question 5 SELECTED RESPONSE — CORRECT: A

Which number is farthest to the LEFT on a number line?

A -4

B -1

C 0

D 2

Farther left means more negative: -4 < -1 < 0 < 2, so -4 is farthest left.

Question 6 SELECTED RESPONSE — CORRECT: C

What is the opposite of -3/4?

- (A) -3/4
- (B) 0
- (C) 3/4
- (D) 4/3

> The opposite of a number is its reflection across 0, so the opposite of -3/4 is 3/4.

Part B — Test-Format Questions

Scoring notes and distractor rationales below each item.

Question 7 TWO-PART QUESTION

Part A — correct answer: C

Between which two consecutive integers does −3.5 fall on a number line?

- (A) −3 and −2
- (B) 3 and 4
- (C) −4 and −3
- (D) −5 and −4

> −3.5 is half a unit to the left of −3, which places it between −4 and −3.

- **A:** Both of those sit to the RIGHT of negative 3.5. Half a unit below negative 3 lands to its left.
- **B:** Those are positive. This number is negative, so it sits left of zero.
- **D:** Both of those sit to the LEFT of negative 3.5.

Part B — correct answer: B

A student says −3.5 is greater than −3 because 3.5 is greater than 3. Why is this reasoning wrong?

- (A) 3.5 is not actually greater than 3.
- (B) On a number line, the further left a number is, the smaller it is, and −3.5 is left of −3.
- (C) Negative numbers cannot be compared.
- (D) Decimals are always smaller than integers.

> For negative numbers, a larger absolute value means a position further LEFT, which means a smaller value. $-3.5 < -3$.

Question 8 WRITTEN RESPONSE · 2 POINTS

Type II Reasoning: Counting the Wrong Direction

Ines plotted −2¼ on a number line. She found −2, then counted one quarter to the RIGHT, landing between −2 and −1.

Explain Ines's misconception and describe where −2¼ actually belongs.

Model answer: −2¼ lies a quarter of a unit to the LEFT of −2, between −3 and −2. Moving right from −2 goes toward zero and gives −1¾, a different number.

2 points	Student explains that on the negative side, larger magnitude means moving LEFT, so −2¼ sits a quarter unit left of −2, between −3 and −2.
1 point	Student says the point belongs between −3 and −2 without explaining the direction.
0 points	Student agrees with Ines or gives an unrelated response.

Part C — Show Your Thinking

Model answer and scoring guidance.

Question 9 WRITTEN RESPONSE · 2 POINTS

Name three different rational numbers between -1 and 0. Write each as both a fraction and a decimal. Explain how you know they are between -1 and 0.

Model answer: Any negative number greater than −1 works, so pick friendly fractions and convert them: −1/2 = −0.5, −1/4 = −0.25, and −3/10 = −0.3. Each is negative, so it sits left of 0, but each has an absolute value less than 1, so it never reaches −1. On the number line all three land inside the single unit between −1 and 0.

Award 2 points for a complete explanation with correct mathematics, 1 point for a correct answer with thin reasoning, 0 for an unrelated response.
